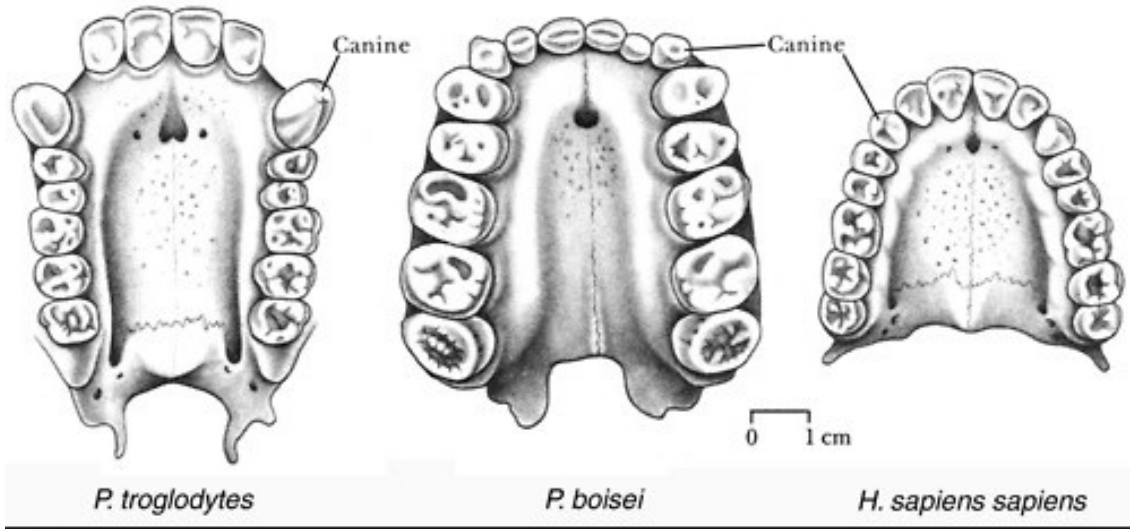
# Hominin Evolutionary Trends




Relative size of cerebral cortex

Prognathism and Dentition

# Relative size of Cerebral Cortex

- Humans – relatively large brains
  - Larger than apes
  - Difference associated with larger cerebral cortex
    - Physically larger
    - Has folds to increase surface area (convolutions)
  - Frontal lobe enlarged
    - Thinking, reasoning, planning, processing
  - Large brain requires large cranium
    - More of skull used for brain in humans than in apes
    - Brow more vertical, without ridges
    - Flatter face, more prominent nose
  - Can see evolutionary trend towards greater cranium size and brain development in extinct human ancestors.

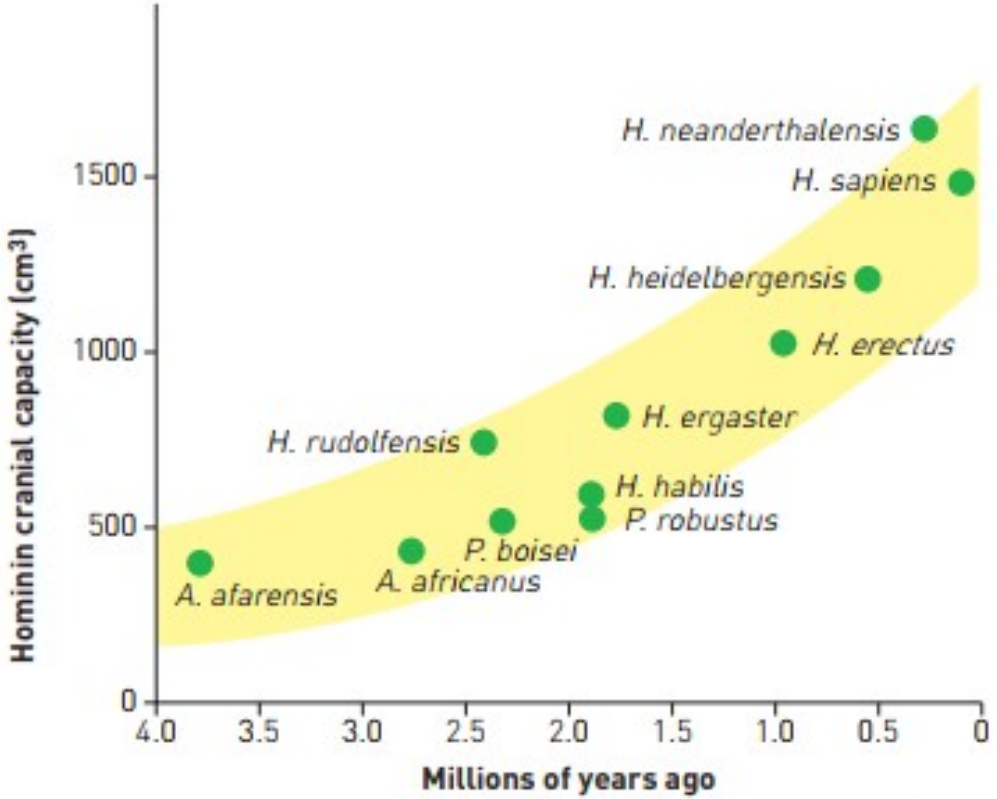



**Figure 18.15** A graph demonstrating the gradual increase in cranial capacity of hominins over time. Average cranial capacity is shown for each species. See Table 18.2 for the full name of each of the species shown.

*Learning Aim: Compare the human cerebral cortex with that of currently living great apes and extinct human ancestors*

# Relative size of Cerebral Cortex

Can see evolutionary trend towards greater cranium size and brain development in hominin ancestors:

- *Australopithecus afarensis* cranium size more similar to modern apes, but has adaptations for bipedalism – indicates brain size increased AFTER bipedalism developed.
- Can't see brain of fossils, but can see cranial capacity, and sometimes **endocasts** showing convolutions on cerebrum (more on this later)
- Gradual increase in relative cranial capacity as more closely related to humans.
- Also foramen magnum positioning, rounding of skull.
- Sometimes only skull fragments are found – can then be difficult to estimate cranial capacity.

**Table 18.2** Hominin cranial capacities

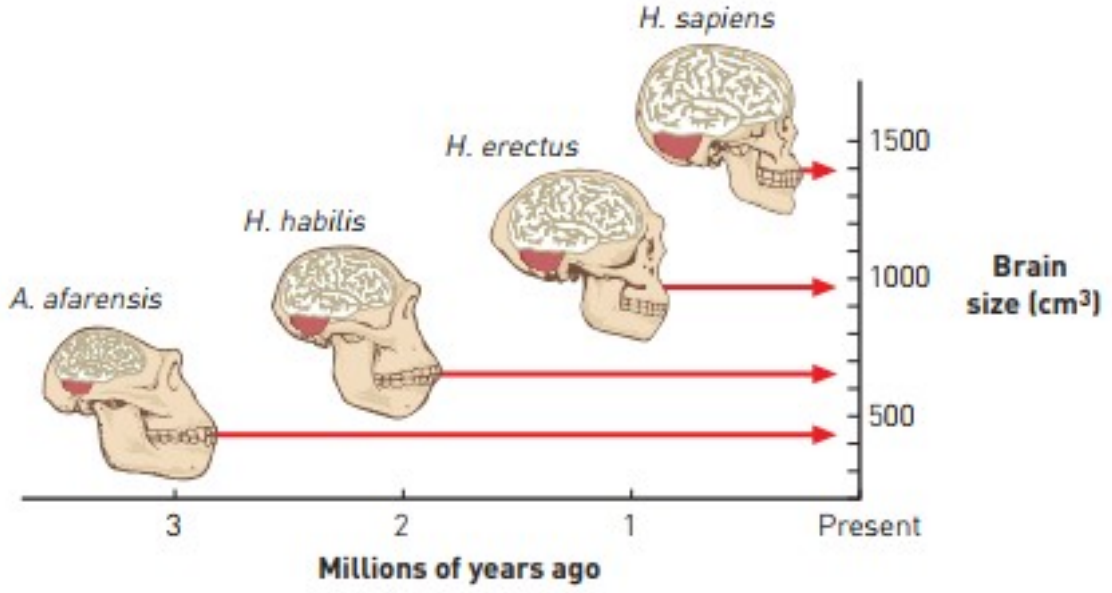
| Hominin | Cranial capacity ($cm^3$) (an estimate of brain size) |
|---|---|
| *Australopithecus afarensis* | 430 |
| *Australopithecus africanus* | 457 |
| *Australopithecus garhi* | 450 |
| *Paranthropus boisei** | 491 |
| *Paranthropus robustus** | 542 |
| *Homo habilis* | 590 |
| *Homo rudolfensis* | 774 |
| *Homo ergaster* | 800 |
| *Homo erectus* | 1004 |
| *Homo heidelbergensis* | 1226 |
| *Homo neanderthalensis* | 1485 |
| *Homo sapiens* | 1350 |

*Learning Aim: Describe evolutionary trends in cranial capacity for extinct human ancestors.*

# Relative size of Cerebral Cortex

- Endocasts
  - Are “casts” of the inside of the skull – see image
  - Provide shape of brain surface
  - May occur naturally, or by scientific casts made of inside of skull
  - Reveal increase in convolutions and relative size of frontal lobe
  - Trend can be seen as hominin species become more recent
  - Even within *H.erectus* –increase seen between earlier and later fossils






*Learning Aim: Describe evolutionary trends in cranial capacity for extinct human ancestors.*

# Prognathism and Dentition

- Prognathism:  forward jutting jaw
- Dentition:  size, shape and position of teeth

## Dentition:

- Humans very distinctive compared to other primates
- Canines do not protrude or interlock – small canine teeth, no diastema
- Parabolic rather than U shaped dental arcade
- Change in dental arcade shape is a trend among hominin ancestors:
  - *A. afarensis* lower jaw and face ape-like
  - *H. habilis*  molars and premolars smaller and narrower, still prominent canines
  - *H. erectus* smaller and narrower, canines smaller
  - Early  *H. sapiens* had larger teeth than modern humans

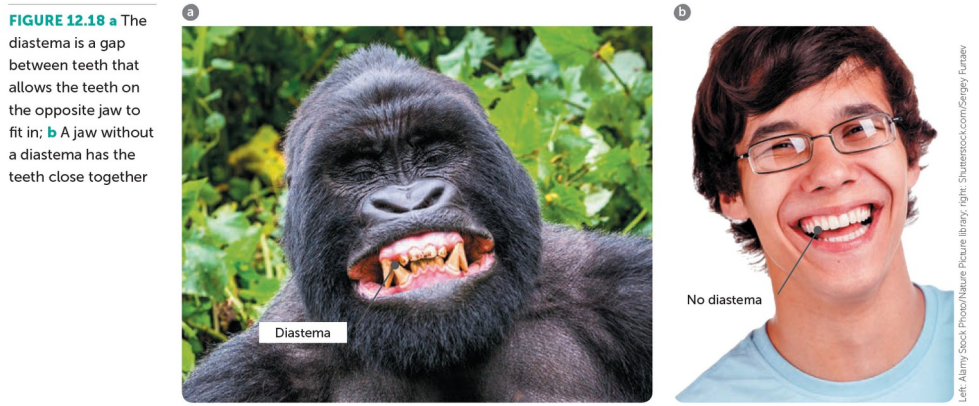



**FIGURE 12.18 a** The diastema is a gap between teeth that allows the teeth on the opposite jaw to fit in; **b** A jaw without a diastema has the teeth close together

Left: Alamy Stock Photo/Nature Picture library; right: Shutterstock.com/Sergey Furtaev

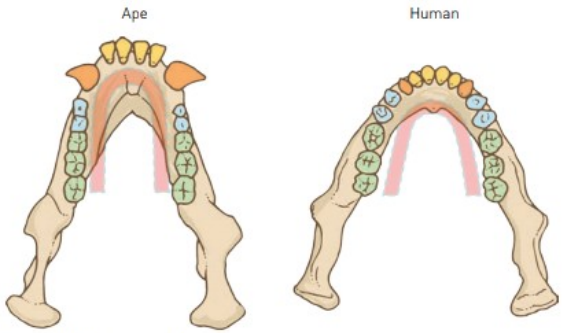
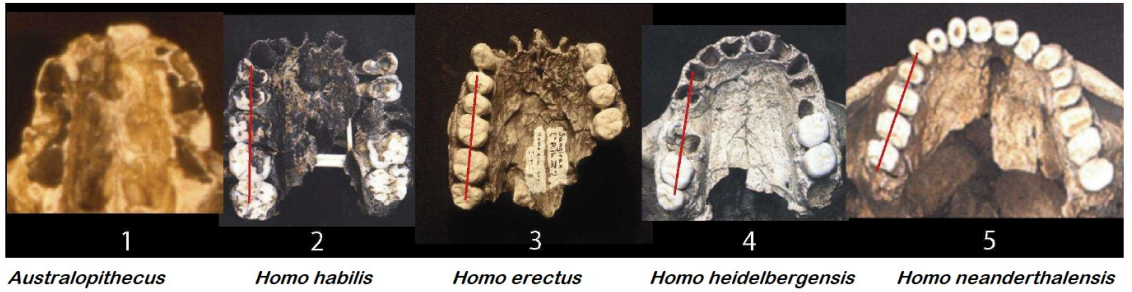



**Figure 18.17** The dental arcade of an ape and a human



**Figure 18.18** A fossil skull of *Homo habilis*




*Learning Aim: Describe evolutionary trends in dentition and dental arcade for extinct human ancestors.*

# Prognathism and Dentition

## Prognathism and Brow Ridge

- Apes and early hominins – forward jutting jaw (prognathism) and distinct brow ridge.
- Evolutionary trend to flatter face, development of chin and more prominent nose.
- Gradual enlargement of frontal lobe led to more distinct forehead, reduction in brow ridge.
- Smaller teeth due to dietary changes led to smaller jaw and flatter face – also provides balance for bipedalism.

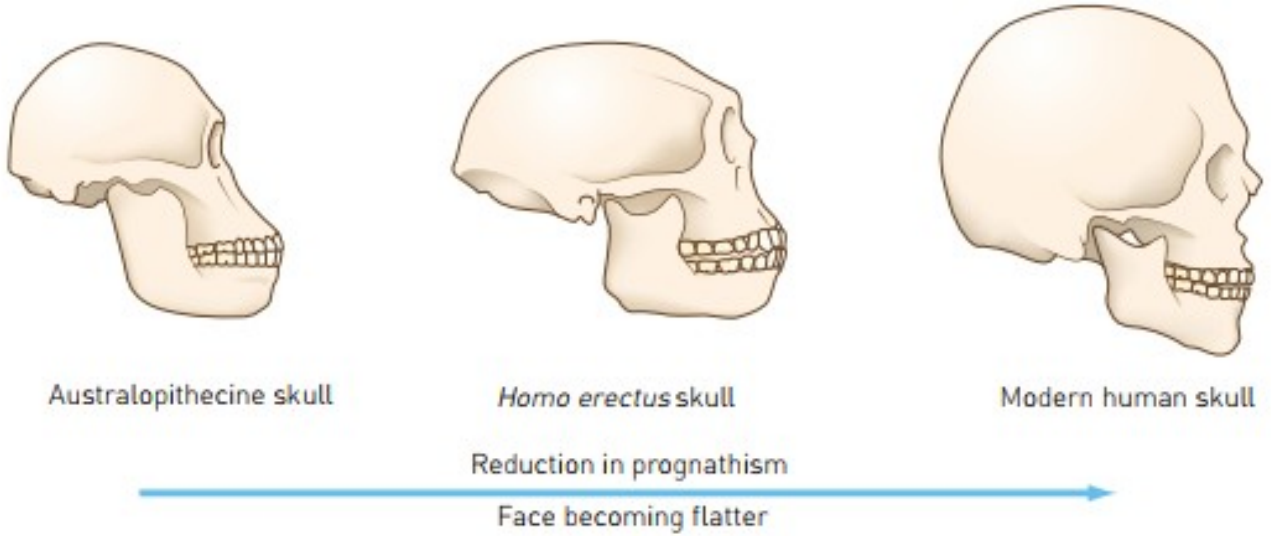



**Figure 18.19** The evolutionary trend towards a flatter face

*Learning Aim: Describe evolutionary trends in prognathism, brow ridge and tooth size for extinct human ancestors.*

**Figure 18.19** The evolutionary trend towards a flatter face

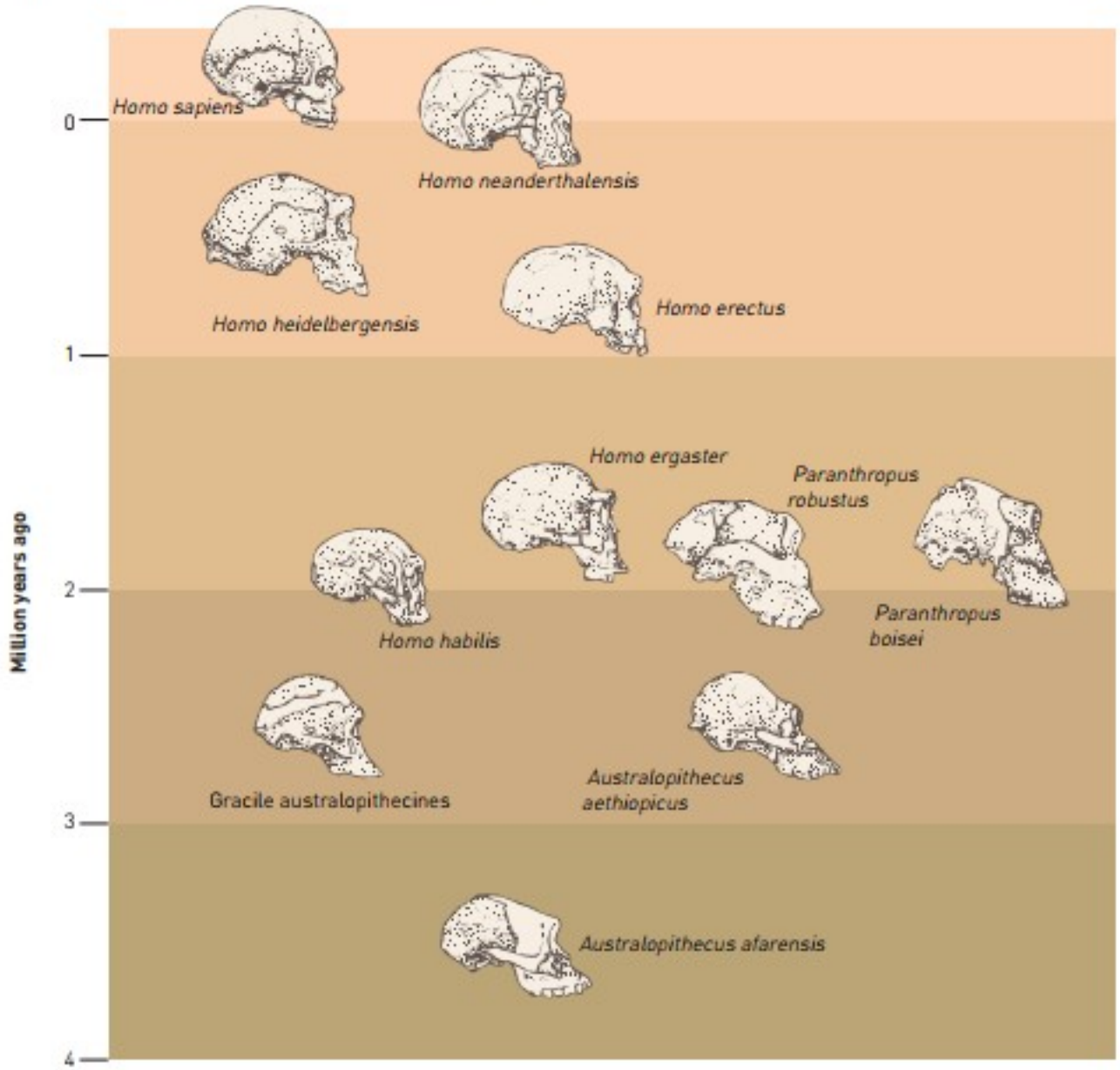



**Figure 18.20** Fossil hominin skulls located in a time frame to indicate the period during which they lived. Note the reduction in prognathism from the early australopithecines to modern humans.